NOTRE DAME UNIVERSITY

BANGLADESH

EEE Lab Report-01

Course Title: Electrical Circuit Analysis Lab

Course Code: EEE-2108

Experiment No: 01

Name of the Experiment: I-V Characteristics of P-N Junction Diode.

Date of Experiment: 20-07-23

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Submitted by:

Group: E

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Submitted To:

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Lecturer, NDUB

Experiment No 1:

Name of the experiment: I-V Characteristics of P-N Junction Diode.

Theory: A p-n junction diode, a non-linear device, conducts only in one direction. The I-V characteristics of the diode is curve between current through the daiode and voltage across the diode. when external voltage is zero, circuit is open & the potential barrier does not allow the current to flow. Therefore, the circuit current is zero. when p-type (anode) is connected to +ve terminal & n-type (cathode) is connected to -ve terminal of the supply voltage, it is known as forward bias. The potential barrier is reduced when diode is in the forward bias condition. At some forward voltage, the potential barrier is altogether eliminated & current starts flowing through the diode. The diode is said to be on state. The current increases with increasing forward voltage.

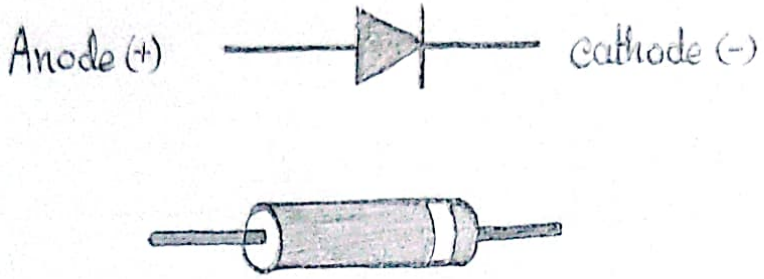

Fig 1.1 : Symbol & Practical view of a Diode

Equipments & Components:

1. p-n junction diode (1N4007) — 1 piece
2. Resistor (1kΩ) — 1 piece
3. Signal generator — 1 unit
4. Trainer Board — 1 unit
5. DC Power Supply — 1 unit
6. Digital Storage Oscilloscope — 1 unit
7. Digital Multimeter — 1unit
8. Chords & wire — as required.

Circuit Diagrams:

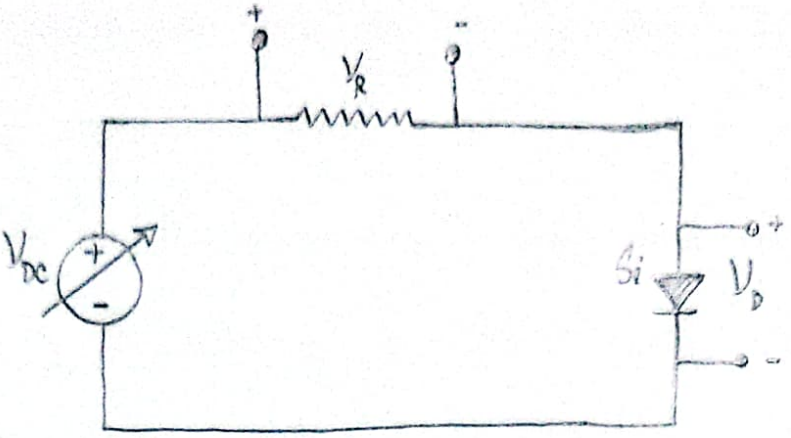

Fig: 1.2: Circuit diagrame for diode forward bias characteristics.

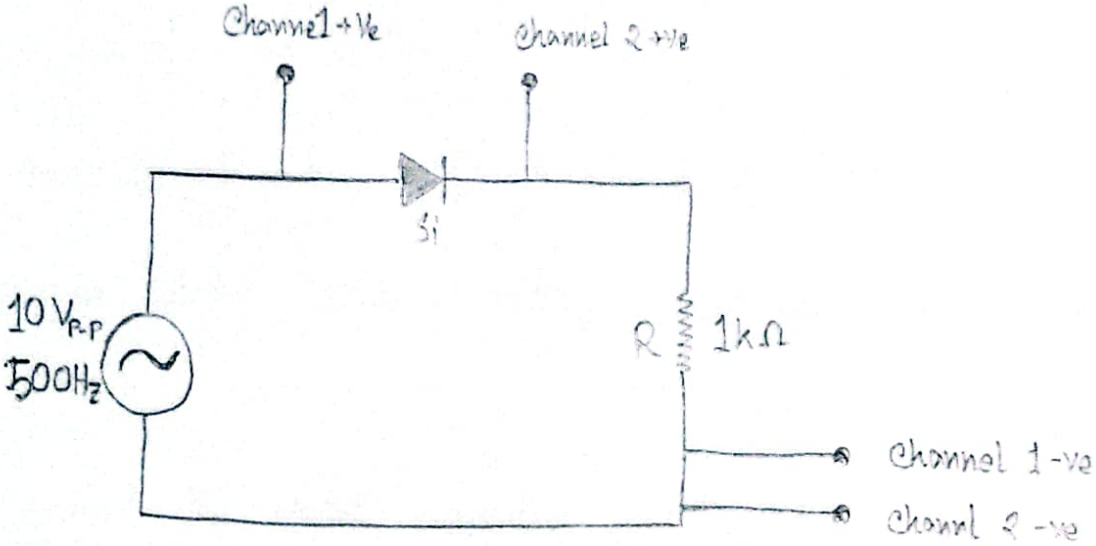

Fig 1.3: Circuit diagram for obtaining diode characteristics from oscilloscope.

Procedures:

1. Connections are made as per the circuit diagram in Fig 1.2 using conventional diode.
2. Switch on the power supply. Vary input voltage (V_{DC}) & measure V_D, V_R for values of $V_D = 0.0, 0.3, 0.6, 0.9, 1.2$ & so on. Obtain maximum value of V_D without increasing V_{DC} beyond 20V.
3. Note down the corresponding current flowing through the diode (I_D) which can be measured by measuring resistance voltage (V_R) & using $I_D = V_R / R$
4. The reading of voltage (V_D) & current (I_D) are tabulated.
5. Graph is plotted between voltage V_D (x-axis) & Current I_D (y-axis)
6. Now connect the circuit according to Fig 1.3. set the Digital Storage Oscilloscope (DSO) in X-Y mode. You will be able to see the I-V Characteristics curve in the DSO.

Group- E

V_{dc}	V_d (V)	I_d (mA)	$I_d' = \frac{V_{dc} - V_d}{R}$
0	0	0	0
0.3	0.226V	0	0.074
0.6	0.468	0.01	0.12
0.9	0.553	0.25	0.347
1.2	0.58	0.50	0.619
1.5	0.602	0.82	0.898
1.8	0.612	1.05	1.188
2.1	0.625	1.39	1.475
2.4	0.634	1.70	1.766
2.7	0.638	1.93	2.062
3	0.646	2.24	2.354
3.3	0.651	2.47	2.649
3.6	0.656	2.81	2.944
3.9	0.659	3.09	3.4241
4.2	0.663	3.36	3.537
4.5	0.668	3.68	3.832
4.8	0.672	4.03	4.128
5.1	0.674	4.27	4.426
5.4	0.678	4.62	4.722
5.7	0.679	4.78	5.021

20/07/23

Student ID: 2, 5, 9, 17R

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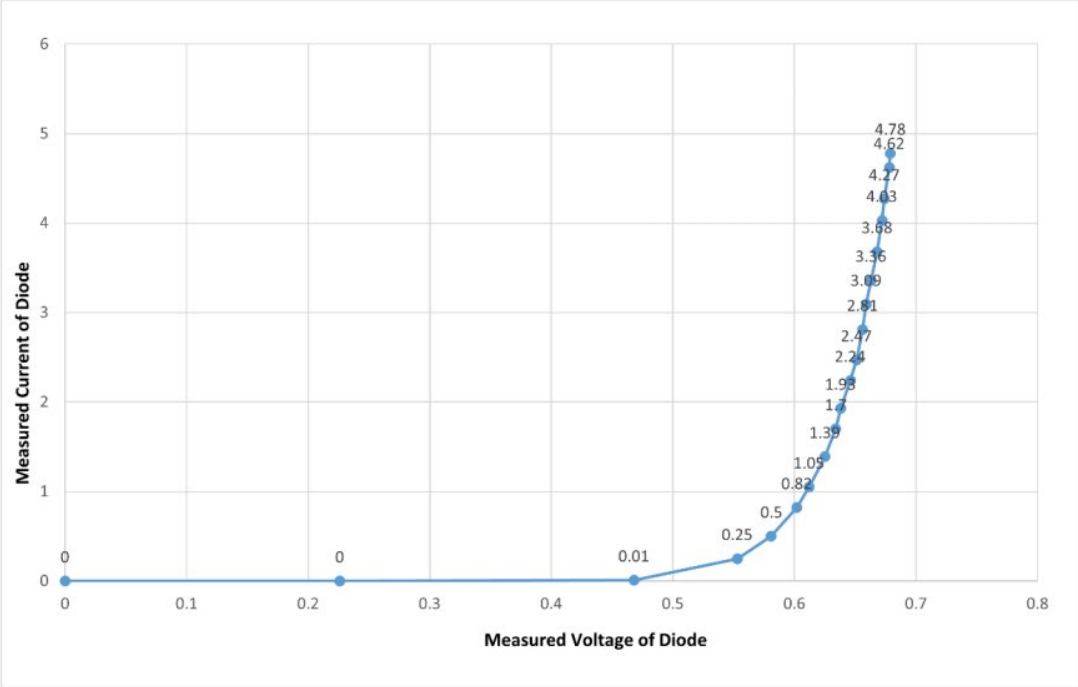
DATA SHEET:

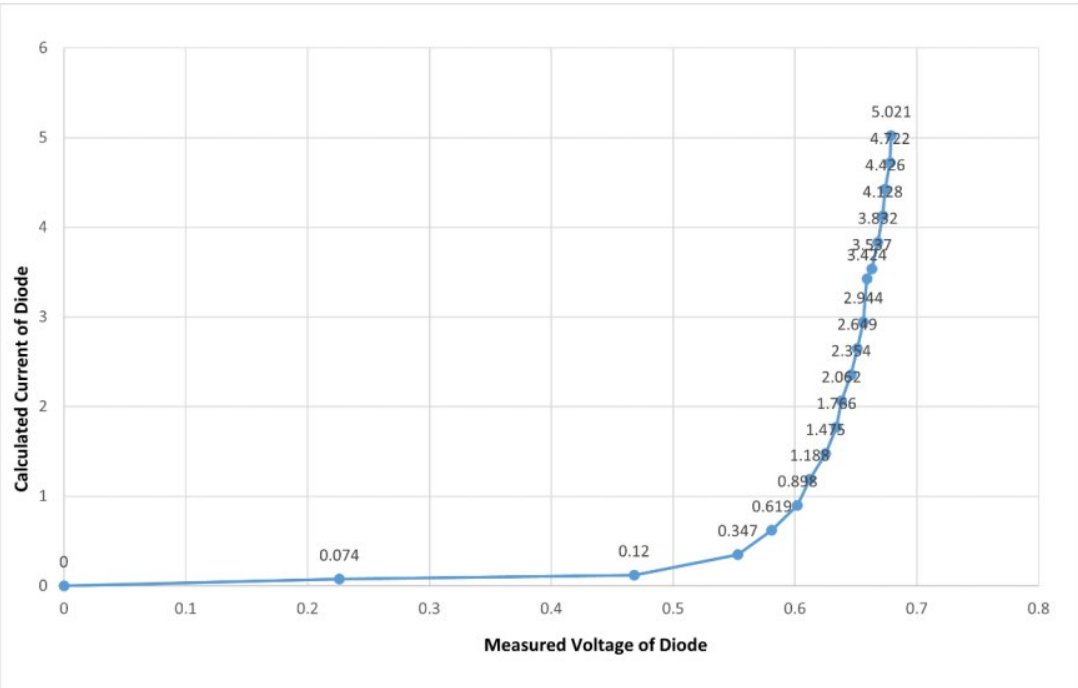
Table 1: Measuring current through the Diode.

Serial no.	DC Supply Voltage (V_{DC}) (V)	Measured Voltage of Diode (V_D) (V)	Measured Current of Diode (I_D) (mA)	Calculated Current of Diode $I = \frac{V_{dc} - V_d}{R}$ (mA)
1.	0	0	0	0
2.	0.3	0.226	0.00	0.074
3.	0.6	0.468	0.01	0.12
4.	0.9	0.553	0.25	0.347
5.	1.2	0.581	0.50	0.619
6.	1.5	0.602	0.82	0.898
7.	1.8	0.612	1.05	1.188
8.	2.1	0.625	1.39	1.475
9.	2.4	0.634	1.70	1.766
10.	2.7	0.638	1.93	2.062
11.	3	0.646	2.24	2.354
12.	3.3	0.651	2.47	2.649
13.	3.6	0.656	2.81	2.944
14.	3.9	0.659	3.09	3.424
15.	4.2	0.663	3.36	3.537
16.	4.5	0.668	3.68	3.832
17.	4.8	0.672	4.03	4.128
18.	5.1	0.674	4.27	4.426
19.	5.4	0.678	4.62	4.722
20.	5.7	0.679	4.78	5.021

Expected Waveforms:



Graph 1.1: I_D vs V_D



Graph 1.2: $\frac{V_{dc}-V_d}{R}$ vs V_D

I-V Characteristics Curve:

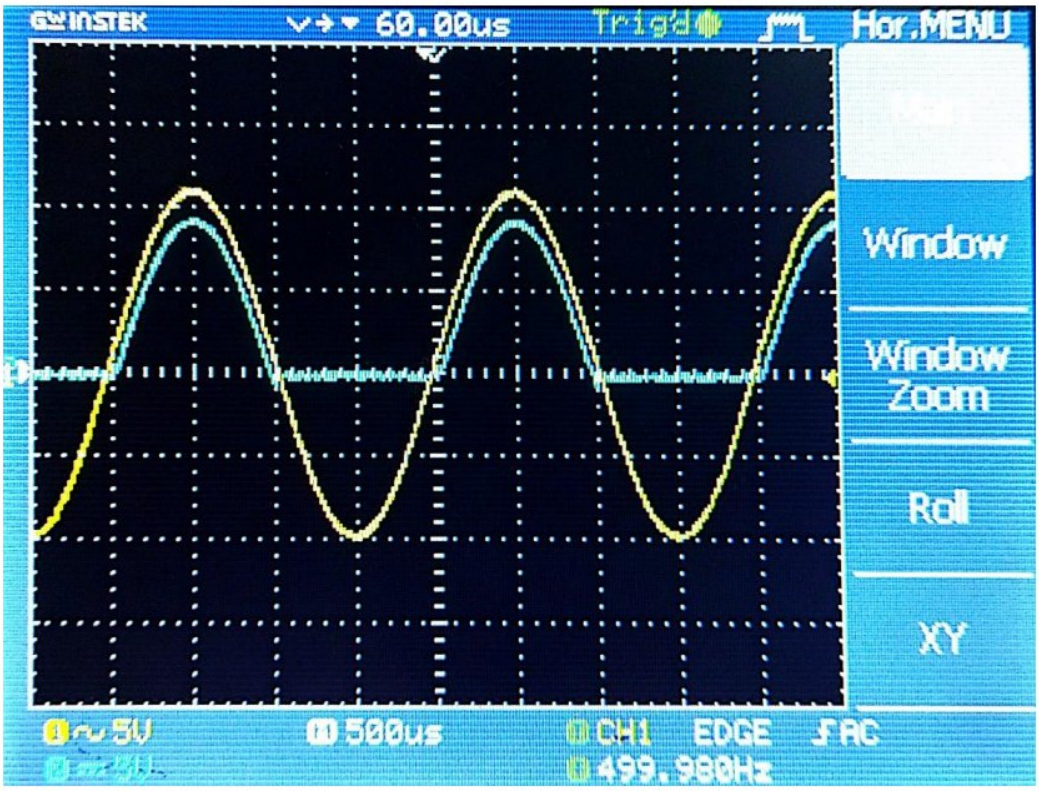


Figure 1.4: Half wave Rectifier

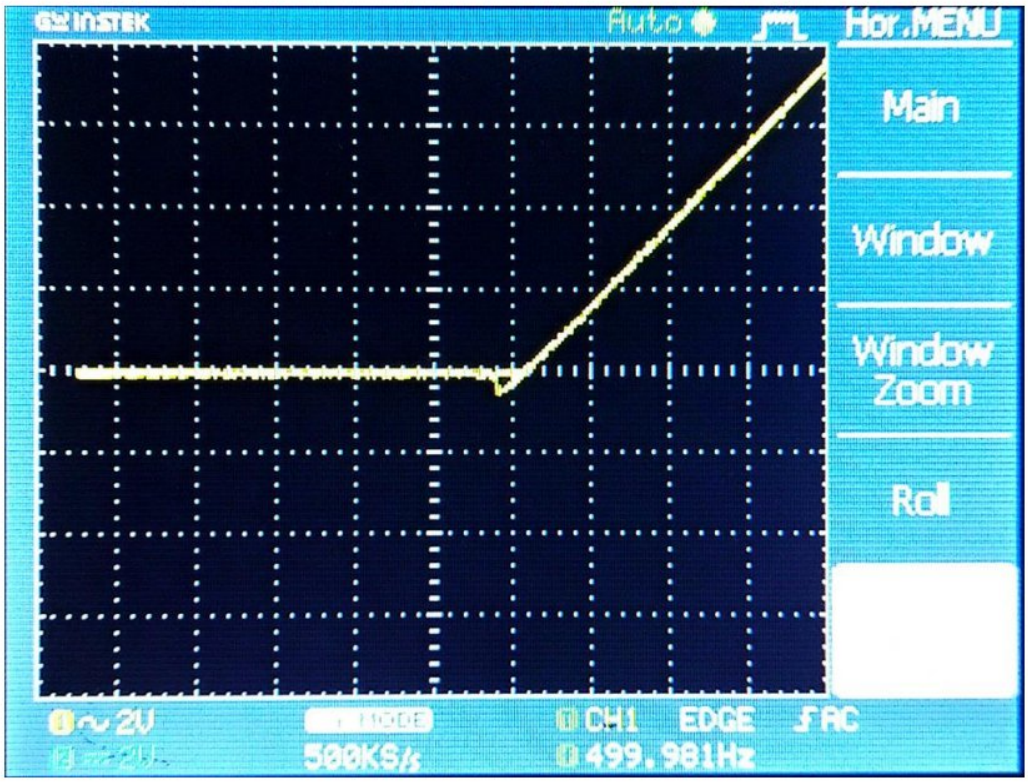


Figure 1.5: I-V Characteristic Curve in X-Y mode

Discussion:

1. Calibrate the oscilloscope for accuracy.
2. Use low voltage & current.
3. Be continuous with reverse voltage to prevent damage.
4. Measure voltage in parallel & when measuring current, short the circuit.
5. Avoid overloading & short-circuit.
6. Follow guidelines & manual.